Do Hidden Markov Regimes Add Out-of-Sample Value?

Unsupervised state discovery, volatility forecasting, and risk-targeting on the S&P 500

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Abstract

Hidden Markov models (HMMs) occupy a storied place in quantitative finance: the Baum–Welch algorithm was co-invented by Leonard Baum, Jim Simons' first partner at Renaissance Technologies. We ask a deliberately sceptical question: on daily S&P 500 data, does a Gaussian HMM add *out-of-sample* value over embarrassingly simple alternatives? We find a split verdict. As a *descriptive* device the HMM is impressive: fitted unsupervised on raw returns, it recovers three economically meaningful states—a calm state that drifts up (+9 bp/day, 9% annualised volatility) and a turbulent state that drifts down (−12 bp/day, 42% volatility)—thereby discovering the leverage effect without being shown a volatility feature. As a *forecasting* device, however, it disappoints: in an expanding-window walk-forward over 3,770 out-of-sample days, the one-parameter RiskMetrics EWMA significantly outperforms the HMM's predictive mixture at next-day volatility forecasting (QLIKE loss, Diebold–Mariano $t = 2.0$), and also beats rule-based regime forecasts and rolling variance. A 10%-volatility-targeting overlay net of costs improves the Sharpe ratio over buy-and-hold for *every* forecaster (0.38 to as high as 0.51) and roughly halves maximum drawdown—but the simplest forecasters deliver the largest gains. The value lies in the idea of volatility targeting, not in sophisticated volatility estimation. Sophistication, by itself, is not edge.

1 Introduction

The romance of regime-switching models in finance begins with a biographical fact: before founding Renaissance Technologies, Jim Simons partnered with Leonard Baum, co-inventor of the algorithm (Baum et al., 1970) that made hidden Markov models estimable. The intellectual bet—that markets, like speech, are driven by unobservable states that can be inferred from noisy observations—has attracted researchers ever since (Hamilton, 1989; Rydén et al., 1998; Ang & Bekaert, 2002; Guidolin & Timmermann, 2007; Nystrup et al., 2018).

This paper asks the question an allocator (or a sceptical PhD committee) would ask: *does the machinery pay its way out of sample?* We compare a from-scratch Gaussian HMM against three deliberately humble alternatives on twenty years of daily S&P 500 data, under a protocol designed to preclude hindsight: expanding-window walk-forward estimation, quarterly refits, an online filter so every forecast at time t uses only information through t, robust loss functions, and formal forecast-comparison tests.

Our contribution is not a new model but an honest accounting, in the spirit of Hansen & Lunde (2005), who famously asked whether anything beats a GARCH(1,1). Our answer

*apati077@ucr.edu. Code, data and every number in this paper are reproducible from `https://github.com/jarvisss007/regime-monte-carlo` (`run_hmm_study.py`); the test suite re-verifies key results in CI.

parallels theirs: for next-day volatility, nothing in our regime-switching toolkit beats a 1994-vintage exponentially weighted moving average (RiskMetrics, 1996).

2 Models

2.1 Gaussian hidden Markov model

Daily returns r_t are modelled as draws from one of $K = 3$ Gaussian emission distributions $\mathcal{N}(\mu_k, \sigma_k^2)$, selected by a latent first-order Markov chain s_t with transition matrix A. Parameters (π, A, μ, σ) are estimated by Baum–Welch (expectation–maximisation over the scaled forward–backward recursions), implemented from scratch in `src/hmm.py`; the most likely state path is decoded with the Viterbi algorithm. Initialisation is deterministic (quantile splits of $|r_t|$), and states are relabelled by ascending σ_k so state 0 is always the calmest. Correctness is verified on synthetic data with *known* hidden states before any market data is touched: the implementation recovers the true path with $> 85\%$ Viterbi accuracy and the true σ_k within 15%.

For out-of-sample prediction we maintain the filtered distribution $p_t = P(s_t \mid r_{1:t})$ with a recursive filter. The predictive variance of tomorrow's return under the mixture $q = p_t A$ is

$$\widehat{\mathrm{Var}}_t(r_{t+1}) \;=\; \sum_k q_k(\sigma_k^2 + \mu_k^2) \;-\; \Big(\sum_k q_k \mu_k\Big)^2. \tag{1}$$

2.2 Comparators

Rule-based regimes. Days are classified calm/normal/turbulent by terciles of trailing 21-day realised volatility (edges estimated on training data only); the forecast is the training-sample variance of the current regime's return pool. This is the observable-state analogue of the HMM. **EWMA.** $\hat{v}_{t+1} = 0.94\,\hat{v}_t + 0.06\,r_t^2$, the RiskMetrics standard (RiskMetrics, 1996). **Rolling.** 21-day trailing variance.

3 Data and protocol

Data are daily S&P 500 adjusted closes, January 1999–December 2018 ($T = 5{,}030$ returns), as bundled in the open-source `arch` package; no vendor access is required. The first 1,260 days (five years) form the initial training window. The HMM and the rule-based regime parameters are re-estimated every 63 trading days on an expanding window; between refits the HMM filter is updated daily. Every forecast of day $t{+}1$ volatility is made strictly with information through day t. This yields 3,770 out-of-sample forecasts per method.

Forecasts are scored with the QLIKE loss $L_t = \log \hat{v}_t + r_t^2/\hat{v}_t$, which is robust to noisy volatility proxies (Patton, 2011). Differences are tested with Diebold–Mariano statistics using a Bartlett/HAC variance with five lags (Diebold & Mariano, 1995).

4 Results

4.1 What the HMM discovers

Table 1 reports the full-sample fit. The states are not mere volatility buckets: mean returns fall monotonically as volatility rises. The calm state combines a +9.2 bp daily drift with 8.7%

Table 1: Three-state Gaussian HMM, S&P 500 daily returns, 1999–2018.

state	μ_k (bp/day)	σ_k (ann.)	self-persistence	E[duration]
calm	+9.2	8.7%	0.979	48.5 days
normal	−1.8	18.5%	0.973	37.4 days
turbulent	−11.5	42.2%	0.969	31.8 days

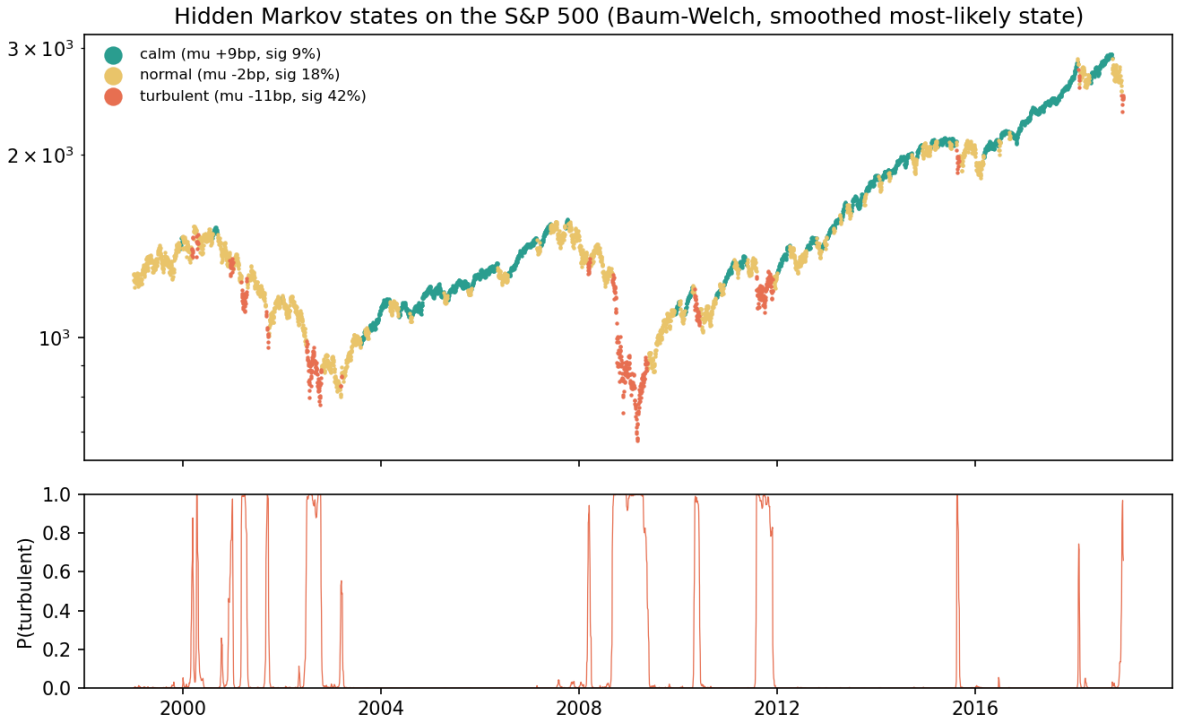


Figure 1: Top: S&P 500 (log scale) coloured by decoded hidden state. Bottom: smoothed probability of the turbulent state.

annualised volatility; the turbulent state combines a −11.5 bp drift with 42.2% volatility. The model was never shown a volatility feature or a sign asymmetry—it inferred the leverage effect from raw returns, consistent with Rydén et al. (1998). All states are highly persistent (expected durations of 32–49 trading days), and the chain never transits directly between calm and turbulent: panic arrives, and recedes, through the intermediate state. Decoded states agree with the rule-based labels on 60% of days, so the hidden states substantially—but not entirely—rediscover observable volatility regimes.

4.2 The forecasting shoot-out

Table 2 contains the paper's central result. The RiskMetrics EWMA achieves the best (lowest) QLIKE, and the Diebold–Mariano tests reject equal accuracy against every regime-based alternative: against the HMM predictive mixture at $t = 2.0$, against rule-based regime volatility at $t = 2.7$, and against rolling variance at $t = 4.1$. A one-parameter filter from 1996 beats the

Table 2: Next-day volatility forecasting, 3,770 out-of-sample days (1999–2018 sample, first five years reserved for initial estimation). QLIKE: lower is better. DM t: positive values mean the EWMA is significantly more accurate.

forecaster	QLIKE	DM t vs EWMA
EWMA(0.94), RiskMetrics	$\mathbf{-8.435}$	—
21-day rolling variance	−8.378	4.12
HMM predictive mixture	−8.357	2.05
rule-based regime vol	−8.350	2.73

Table 3: 10% vol-target overlay on the S&P 500, net of 2 bps/side, same 3,770 out-of-sample days.

overlay driven by	ann. return	Sharpe	max drawdown	realised vol
rolling variance	4.8%	**0.51**	−25.8%	10.0%
EWMA(0.94)	4.7%	**0.51**	−25.1%	9.9%
rule-based regimes	4.6%	0.49	−29.9%	10.2%
HMM predictive mixture	3.7%	0.42	−29.7%	9.7%
buy & hold (no overlay)	5.5%	0.38	−56.8%	18.3%

Baum–Welch machinery at its supposed home game.

The mechanism is not mysterious. Regime forecasts are, by construction, piecewise-constant: within a regime the forecast barely moves, and at regime boundaries it jumps between a small number of levels. Volatility itself evolves continuously. The EWMA's geometric weighting tracks that continuous evolution day by day, while the HMM must express all variation through shifts in a three-point mixture.

4.3 Does forecasting quality cash out? The vol-target overlay

Following the logic of volatility-managed portfolios (Moreira & Muir, 2017), we run a 10%-annualised-volatility-target overlay: the position in the index is $w_t = \min\big(1, \sigma_{\text{tgt}}/\widehat{\sigma}_t\big)$, with 2 bps per side charged on turnover. Table 3 shows that *every* forecaster's overlay improves on buy-and-hold's Sharpe ratio (0.38) and roughly halves its −57% maximum drawdown—and that the ranking follows forecast accuracy: the simplest forecasters (rolling, EWMA, Sharpe 0.51–0.52) beat the HMM overlay (0.42). Each overlay realises close to its 10% volatility target, so the comparison is at matched risk.

5 Discussion

The verdict is a clean split along the description/prediction boundary. As a descriptive and generative device the HMM earns its reputation: it discovers economically interpretable states unsupervised, and (as shown in the companion simulation study in the same repository) regime-conditional resampling reproduces the fat tails and volatility clustering that Gaussian diffusion models miss entirely. As a short-horizon forecasting device it does not: the filter-plus-mixture prediction is dominated by exponential smoothing, echoing Hansen & Lunde (2005) in a regime-

switching key. For the practitioner the actionable finding is Table 3: volatility targeting is robustly valuable, and it does not require—indeed is mildly hurt by—sophisticated state inference. For the researcher, the finding is a caution against equating model elegance with predictive content.

6 Limitations

(i) One asset and one frequency; the HMM's relative standing may differ for multi-asset problems, longer horizons, or applications that exploit the full predictive *distribution* rather than its variance. (ii) Gaussian emissions with $K = 3$; richer emissions (Student-t) or model-selected K could narrow the gap. (iii) Quarterly refits are a compute compromise; daily refits are conceptually identical but were not run. (iv) We did not include a GARCH(1,1) comparator; given Hansen & Lunde (2005), we expect it to perform near the EWMA, which is its integrated special case. (v) The overlay's improvement over buy-and-hold is economic rather than formally tested; our significance claims concern forecast accuracy only.

7 Conclusion

Fitted honestly and evaluated out of sample, a Gaussian hidden Markov model on S&P 500 returns is a very good storyteller and a mediocre forecaster. The states it discovers are real and economically meaningful; the forecasts it produces are significantly worse than a one-parameter EWMA; and the robust practical value in this exercise—volatility targeting—is available to anyone with a rolling window. The Renaissance lesson, at least at this scale and with this data, is not that hidden-state models print money; it is that disciplined evaluation separates what a model *reveals* from what it can *predict*.

Reproducibility

`python run_hmm_study.py` reproduces every table; `python plots_hmm.py` reproduces Figure 1; `pytest` re-verifies the implementation (including known-state recovery on synthetic data) and runs in CI on every commit. Data ship with the `arch` package.

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